

Jake Capmeil

Student-Researcher

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Education

Hunter College High School / Class of 2028

Relevant coursework

AP Computer Science A, Extended-Honors Math (PreCalculus, Geometry, Statistics, Probability), Honors Computer Science, Honors Physics, AP French

Awards

National French Grand Concours, native-speakers division. two-time Gold Medalist.

Research

Threshold Tuning Under Class Imbalance for Predicting Student Preference in AI-Assisted Tutoring / CMU

2026- PRESENT

Modeled student preference for AI versus human tutoring under CMU research mentorship, comparing three classifiers with k-fold cross-validation, MCC optimization, and threshold tuning to address class imbalance. Demonstrated that small, self-collected datasets can support meaningful prediction, as a proof of concept for resource-limited institutions.

Aspect Ratio Effects on Flight Performance at Fixed Planform Area in Free-Flight Gliders / HCHS

2024 - 2025

Engineered a modular sailplane that swaps wings of equal area but different aspect ratios, then ran systematic flight trials to isolate what drives lift, stability, and glide efficiency.

Bathymetric Control of Breaking Wave Height in Shoaling Nearshore Environments / HCHS

2023 - 2024

Designed and built a working wave tank with interchangeable seafloor topographies to quantify how slope, depth, and contour reshape a wave. Measured propagation and shoaling across each model.

Software Projects

Bloom

2026- PRESENT

Shipped a full-stack habit-tracking app on Cloudflare Workers with Google authentication, a Supabase backend, and a built-in Gemini AI coach. Progress renders as a growing tree, translating streak consistency into visible growth.

Khromate

2026 - Present

Built a generative art platform where users produce royalty-free work from coded algorithms, then download it or publish it to a shared in-site gallery.

Academic Track

Carnegie Mellon CS Scholars / CMU

Summer 2026

Fully funded, merit-based residential program with a ~4% acceptance rate. Four weeks of college-level CS and Graph Theory coursework alongside a research project presented at a closing symposium.

Science Research Seminar / HCHS

2025 - PRESENT

Selective lab-prep seminar where students who pass a rigorous qualifying exam analyze publications, cover research ethics and statistics, and take a rigorous ML / Advanced Python course with hands-on projects.

Skills

Programming & Data

Python, pandas, Supervised Learning, Neural Networks, Data Pipelines, Game Development

Full-Stack & Version Control

Supabase, GitHub, Cloudflare, Render, Netlify

AI-Assisted Engineering

Claude Code, Agent Workflows, CLAUDE.md Specs, AI-Assisted Coding & Data Analysis, Full App Launching, AI In-App Implementation

Software

Excel, Word, PowerPoint, Logic Pro, Pixelmator Pro, Adobe Express, Affinity

Music Production

Recording Equipment, DAWs, Sampling, Piano

Languages

English (Fluent), French (Fluent)

Additional Experience

Interact Club of Chinatown

Vice President and Founder, organizing service events and running all facets of club logistics. Raised \$1,500+ in the first year and grew memberships to 40+ students.

English Peer Tutor

Chosen as one of the top writers in the grade to mentor younger students in grammar, structure, analysis, and creative writing.

Cross Country & Track

Year-round varsity athlete who competed at Cities alongside serving as a team leader.

Photography Assistant

On-set production support, shooting/editing promo content, and prepping fine-art prints for gallery clients.

Laru Beya Summer

Nonprofit surf instruction and water safety for underserved youth.